

CLAY MILLENNIUM PROBLEM · RESEARCH PROGRAMME DOCUMENT

Yang–Mills Existence and Mass Gap

Formal Acceptance and Research Specification

Christian Bucher¹

Version 1.0 · 2026-09-24 · Status: draft for author review

<https://github.com/christian140903-sudo/nextool/tree/main/research/yang-mills>

Abstract

This document fixes, in verifiable form, what a resolution of the Clay Millennium Problem “Yang–Mills Existence and Mass Gap” must establish. It also sets out a two-track research programme to pursue one: a constructive track (existence and mass gap) and a foundational-audit track (a rigorous counterexample or a proof of ill-posedness). The official problem description [JW] and the Millennium Prize rules [CMI18] are frozen by SHA-256 fingerprint. From them we derive an acceptance contract of 25 requirements, an interpretation register of 6 points where the official text is not formally specific, an assumption ledger of 33 classified premises and a map of 9 separate transition theorems. A build script validates all of this and computes the closure status of every transition. **This document does not contain a solution.** In the present revision 1 of 9 transitions is closed; this matches the literature, and the problem remains open.

Status at a glance

- **Problem status.** Clay Mathematics Institute lists the problem as *Unsolved* (checked 2026-09-24). No proposed solution satisfies the prize conditions of [CMI18, §4].
- **What this document is.** The fixed yardstick against which every later computation, lemma and manuscript of the programme is measured.
- **What it is not.** A proof, a partial proof or a claim of progress on the mathematics itself.
- **Machine-checked closure.** 1/9 transitions closed. 6 ledger entries are **OPEN**, and 5 tempting shortcuts are recorded as **FALSE** (Section 7).

Contents

1	Purpose, scope and conventions	3
2	Normative sources and target freeze	3
3	Formal restatement	4
3.1	Conventions	4
3.2	Wightman theory of local gauge-invariant fields	5
3.3	Euclidean route: Osterwalder–Schrader	5
3.4	Mass gap and mass	5
3.5	What makes the theory “Yang–Mills”	6
3.6	The main statement, fully quantified	6

¹Drafting assistance: Claude Code, an AI system by Anthropic. Every source statement was checked against the primary documents listed in Section 2 and in the references. In line with the COPE position on authorship and AI tools [COPE23], the AI tool is not an author. Responsibility for the content lies with the author.

4	Acceptance contract	7
5	Interpretation register	8
6	Quantifier ledger	10
7	Assumption ledger	10
8	Transition map and closure status	13
9	State of the art (verified references)	14
10	Track P: construction and mass gap	15
10.1	Choice of representation	15
10.2	Milestones	16
11	Track N: foundational audit and counterexample	16
11.1	What the “questionable foundation” hypothesis must become	16
11.2	Admissible outcomes	16
11.3	What is not a counterexample	17
11.4	Precedents: how foundations were actually corrected	17
11.5	Prior assessment	17
12	Evidence standard and verification	17
12.1	Evidence classes	17
12.2	Adversarial review	18
12.3	Mandatory sanity tests	18
12.4	Formal verification	18
13	Manuscript architecture and traceability	18
13.1	Architecture	18
13.2	Writing rules	18
13.3	Traceability matrix (template)	19
14	Publication, priority and integrity	19
15	Review of the working draft	19
16	Roadmap	20
A	Deutsche Zusammenfassung	21
B	Build and reproducibility	21

1 Purpose, scope and conventions

A Millennium Problem is not solved by a convincing argument, a numerical result or a new theory. It is solved only when the global mathematics community accepts a complete proof, published in a refereed venue, that answers the specific questions of the official description [CMI18, §§4–5, 7]. Most failed attempts do not fail on the central idea. They fail on a mismatch between what was proved and what was asked: a wrong quantifier, a limit that loses a property, a regulator that is never removed, or a theorem used outside its hypotheses.

This specification exists to make that mismatch impossible to overlook. It serves three purposes:

- (i) **Target freeze.** It records the normative texts, verbatim and fingerprinted (Section 2).
- (ii) **Acceptance contract.** It turns these texts into atomic, individually checkable requirements (Sections 3–6).
- (iii) **Research control.** It classifies every premise the programme may use and tracks which transition theorems are closed (Sections 7–8). It also defines the two tracks and the standards of evidence and review (Sections 10–14).

Normative language. The key words MUST, MUST NOT, SHOULD and MAY are used in the sense of BCP 14 [RFC2119, RFC8174]. They bind the research programme. They say nothing about mathematical truth.

Identifiers. *Sn* normative sources; *YM–nn* requirements; *I–n* interpretations; *A–nnn* ledger entries; *Tn* transitions; *Q–n* quantifier rules; *En* findings on the working draft. All identifiers are hyperlinks.

Single source of truth. Tables in Sections 4, 5, 7 and 8 are generated from `ledger.mjs`. `node build.mjs` refuses to build if an identifier is duplicated or dangling, if a mathematical requirement is not covered by a transition, or if a transition rests on a premise whose status is `FALSE`, `HEURISTIC`, `NUMERICAL` or `PREPRINT`. A machine-readable copy is written to `ledger.json` (Appendix B).

2 Normative sources and target freeze

ID Source, retrieval and fingerprint

- S1 A. Jaffe, E. Witten, Quantum Yang–Mills Theory (official problem description)
<https://www.claymath.org/wp-content/uploads/2022/06/yangmills.pdf>
retrieved 2026-09-24, 195,976 bytes, SHA-256 3558403ca14c11e382f73a09e548222708540bfdf478cf96aa11c52d43e23e09
 - S2 Clay Mathematics Institute, Millennium Prize Description and Rules (adopted 2018-09-26)
https://www.claymath.org/wp-content/uploads/2022/03/millennium_prize_rules_0.pdf
retrieved 2026-09-24, 43,263 bytes, SHA-256 9b5003745c0ae7268dc7769f83e1c61eaca674f28631f1df2cd8400976640b2a
 - S3 R. F. Streater, A. S. Wightman, PCT, Spin and Statistics, and All That (1964) — reference [45] of S1
 - S4 K. Osterwalder, R. Schrader, CMP 31 (1973) 83–112 and CMP 42 (1975) 281–305 — reference [35] of S1
-

Hierarchy. S2 governs procedure: eligibility, publication and evaluation. S1 governs mathematical content. S3 and S4 fix the minimum axiomatic strength that S1 requires. When a later document (including this one) deviates from S1 or S2, S1 and S2 prevail.

Freeze rule. If the SHA-256 fingerprint of S1 or S2 at its official URL changes, this specification MUST be re-reviewed in a new revision before any further claim is measured against it.

The problem, verbatim. From S1, §4:

To establish existence of four-dimensional quantum gauge theory with gauge group G , one should define a quantum field theory (in the above sense) with local quantum field operators in correspondence with the gauge-invariant local polynomials in the curvature F and its covariant derivatives, such as $\text{Tr } F_{ij} F_{kl}(x)$. Correlation functions of the quantum field operators should agree at short distances with the predictions of asymptotic freedom and perturbative renormalization theory, as described in textbooks. Those predictions include among other things the existence of a stress tensor and an operator product expansion, having prescribed local singularities predicted by asymptotic freedom.

[...] A quantum field theory has a mass gap Δ if H has no spectrum in the interval $(0, \Delta)$ for some $\Delta > 0$. The supremum of such Δ is the mass m , and we require $m < \infty$.

Yang–Mills Existence and Mass Gap. *Prove that for any compact simple gauge group G , a non-trivial quantum Yang–Mills theory exists on $\mathbb{R}^4$ and has a mass gap $\Delta > 0$. Existence includes establishing axiomatic properties at least as strong as those cited in [45, 35].*

S1, footnote 2: “We specifically exclude weak-existence (compactness) as the solution to the existence part of the Millennium problem, unless one also uses other techniques to establish properties of the limit (such as the existence of a mass gap and the axioms).”

Counterexamples, verbatim. From S2, §5(b)–(c): “In the case of the P versus NP Problem and the Navier-Stokes Problem, a resolution in either direction will be evaluated by the standard evaluation procedure [...]. In the case of the other Problems, if a counterexample is proposed, such counterexample shall be evaluated using the same evaluation procedure as if it were a Proposed Solution. (i) If, in the opinion of CMI, the counterexample effectively resolves the Problem, then CMI may recommend the award of a Prize. (ii) If, alternatively, the counterexample shows that the original Problem survives after reformulation or elimination of some special case, then CMI may recommend that a small prize [...] be awarded to the author. The money for this prize will not be taken from the Problem fund, but from other CMI funds.”

3 Formal restatement

3.1 Conventions

Minkowski space $\mathbb{R}^{1,3}$ has metric signature $(+, -, -, -)$. $\bar{V}_+ = \{p \in \mathbb{R}^{1,3} : p^0 \geq |\vec{p}|\}$ is the closed forward cone. $\mathcal{P}_+^\uparrow$ is the restricted Poincaré group. $\mathcal{S}(\mathbb{R}^{4n})$ is the Schwartz space. Euclidean space-time is $\mathbb{R}^4$ with Euclidean group $E(4)$ and time reflection $\Theta : (x^0, \vec{x}) \mapsto (-x^0, \vec{x})$. $\mathfrak{g}$ denotes a compact simple Lie algebra and G a compact connected Lie group with $\text{Lie}(G) = \mathfrak{g}$ (see l-2).

3.2 Wightman theory of local gauge-invariant fields

Definition 3.1 (Target structure). A *Wightman theory of local gauge-invariant fields* is a tuple $\mathcal{T} = (\mathcal{H}, U, \Omega, \mathcal{D}, \{\mathcal{O}_i\}_{i \in I})$ such that

- (W1) $\mathcal{H}$ is a separable Hilbert space and U a strongly continuous unitary representation of $\mathcal{P}_+^\uparrow$ (or its covering group) on $\mathcal{H}$. YM-05
- (W2) The joint spectrum of the generators $(H, \vec{P})$ of translations lies in $\bar{V}_+$. YM-06
- (W3) There is a vector Ω with $U(g)\Omega = \Omega$ for all g , unique up to a phase. YM-07
- (W4) Each $\mathcal{O}_i$ is an operator-valued tempered distribution. The operators $\mathcal{O}_i(f)$, $f \in \mathcal{S}(\mathbb{R}^4)$, are defined on a common dense domain $\mathcal{D} \ni \Omega$ with $U(g)\mathcal{D} \subseteq \mathcal{D}$ and $\mathcal{O}_i(f)\mathcal{D} \subseteq \mathcal{D}$. The fields transform covariantly under U , and Ω is cyclic for the polynomial algebra of the $\mathcal{O}_i(f)$. YM-08
- (W5) If $\text{supp } f$ and $\text{supp } g$ are space-like separated, then $[\mathcal{O}_i(f), \mathcal{O}_j(g)] = 0$ on $\mathcal{D}$. YM-09

The gauge-invariant local fields of pure Yang–Mills theory are bosonic, so (W5) uses commutators only.

3.3 Euclidean route: Osterwalder–Schrader

Definition 3.2 (OS axioms, in the corrected form of [OS75]). A family of Schwinger functions $\{S_n\}$ of the local fields satisfies

- (OS0) *Regularity with linear growth* (OS0'). The S_n are distributions on test functions vanishing at coinciding points. Their order grows at most linearly in n and their constants at most factorially (schematically $|S_n(f)| \leq \alpha(n!)^\beta |f|_{sn}$; the exact form is in [OS75]).
- (OS1) *Euclidean covariance* under $E(4)$.
- (OS2) *Reflection positivity*: $\sum_{n,m} S_{n+m}(\Theta \bar{f}_n \otimes f_m) \geq 0$ for all finite families f_n supported at positive times.
- (OS3) *Symmetry* under permutations of the arguments.
- (OS4) *Cluster property*.

Imported theorem 3.3 (OS reconstruction; A-001). *Schwinger functions satisfying OS0' and OS1–OS3 are the analytic continuations of the Wightman functions of a theory satisfying (W1)–(W5), except possibly the uniqueness in (W3). The cluster property OS4 is equivalent to uniqueness of the vacuum.*

Remark 3.4 (A precedent for the foundational audit). OS I [OS73] claimed reconstruction without the linear growth condition. The proof contained an error (its Lemma 8.8), and OS II [OS75] repaired the theorem by adding OS0' (A-002). The requirement of S1, “at least as strong as those cited in [45, 35]”, therefore points to the *corrected* theory. A construction that verifies only OS0 is incomplete.

3.4 Mass gap and mass

Definition 3.5 (S1 §4). $\mathcal{T}$ has a *mass gap* $\Delta > 0$ if $\sigma(H) \cap (0, \Delta) = \emptyset$. The *mass* is $m := \sup\{\Delta > 0 : \sigma(H) \cap (0, \Delta) = \emptyset\} \in (0, \infty]$, and S1 requires $m < \infty$.

Lemma 3.6. *Under (W1)–(W3), $m < \infty \iff H \neq 0$. If moreover every translation-invariant vector is a multiple of Ω (A-027), then $m < \infty \iff \mathcal{H} \neq \mathbb{C}\Omega$.*

Proof. $H \geq 0$ by (W2). If $H \neq 0$, some $E > 0$ lies in $\sigma(H)$, and every admissible Δ satisfies $\Delta \leq E$, so $m \leq E < \infty$. If $H = 0$, then $\sigma(H) = \{0\}$ and every $\Delta > 0$ is admissible, so $m = \infty$. For the second claim: if $H = 0$, then (W2) forces $\vec{P} = 0$. Every vector is then translation-invariant, hence a multiple of Ω . $\square$

Remark 3.7 (Equivalent forms and a consistency check). Under (W1)–(W5) with a unique vacuum, a gap of H is equivalent to a gap of the mass operator $M = \sqrt{H^2 - \vec{P}^2}$ on $\Omega^\perp$. It is also equivalent to uniform exponential decay in Euclidean time of truncated two-point functions of a vacuum-cyclic family of fields (A-016). S1 §5 records the consequence (A-017): for every $C < \Delta$ and every local $\mathcal{O}$ with $\langle \Omega, \mathcal{O} \Omega \rangle = 0$,

$$|\langle \Omega, \mathcal{O}(\vec{x}) \mathcal{O}(\vec{y}) \Omega \rangle| \leq e^{-C|\vec{x}-\vec{y}|} \quad \text{for } |\vec{x} - \vec{y}| \text{ sufficiently large.}$$

The hypothesis $\langle \Omega, \mathcal{O} \Omega \rangle = 0$ is essential: without it the left side tends to $|\langle \Omega, \mathcal{O} \Omega \rangle|^2 \neq 0$.

3.5 What makes the theory “Yang–Mills”

Classically, a connection A on a principal G -bundle has curvature $F = dA + A \wedge A$, and the action is $\frac{1}{4g^2} \int \text{Tr } F \wedge *F$ (S1, eq. (1)). The classical equations describe *massless* waves. The gap is a purely quantum phenomenon. What distinguishes the quantum theory from an arbitrary massive theory is its short-distance behaviour (YM-11, YM-12), which is governed by the renormalization group:

$$\mu \frac{dg}{d\mu} = \beta(g) = -b_0 g^3 - b_1 g^5 + O(g^7), \quad b_0 = \frac{11}{3} \frac{C_A}{16\pi^2}, \quad b_1 = \frac{34}{3} \frac{C_A^2}{(16\pi^2)^2},$$

with $C_A = h^\vee(\mathfrak{g})$ the dual Coxeter number [GW73, Pol73, Cas74, Jon74]. To leading order the running coupling is $\bar{g}^2(\mu) = 1/(b_0 \ln(\mu^2/\Lambda^2))$. Here Λ is the renormalization-group-invariant scale: the classical scale invariance is broken, and every mass of the quantum theory is expected to be a pure number times Λ , i.e. $m = c_g \Lambda$ (an expectation, **HEURISTIC**).

$\mathfrak{g}$	A_n	B_n	C_n	D_n	E_6	E_7	E_8	F_4	G_2
range	$n \geq 1$	$n \geq 2$	$n \geq 3$	$n \geq 4$					
$h^\vee(\mathfrak{g})$	$n+1$	$2n-1$	$n+1$	$2n-2$	12	18	30	9	4

Table 1: The compact simple Lie algebras and their dual Coxeter numbers (A-028). Since $h^\vee > 0$ in every case, $b_0 > 0$: pure Yang–Mills is asymptotically free for every $\mathfrak{g}$ in the scope of YM-02.

3.6 The main statement, fully quantified

Statement 3.8 (MAIN).

$$\forall \mathfrak{g} \quad \exists \mathcal{T} : \underbrace{\text{Ax}(\mathcal{T})}_{\text{YM-05–09, 15}} \wedge \underbrace{\text{Fields}_{\mathfrak{g}}(\mathcal{T})}_{\text{YM-10}} \wedge \underbrace{\text{UV}_{\mathfrak{g}}(\mathcal{T})}_{\text{YM-11, 12}} \wedge \underbrace{\text{NT}(\mathcal{T})}_{\text{YM-04}} \wedge \underbrace{\exists \Delta > 0 : \sigma(H) \cap (0, \Delta) = \emptyset}_{\text{YM-13}} \wedge \underbrace{m(\mathcal{T}) < \infty}_{\text{YM-14}}$$

with $\mathcal{T}$ on $\mathbb{R}^4$ (YM-03) and existence not by compactness alone (YM-16).

Statement 3.9 (Negation).

$$\neg \text{MAIN} \iff \exists \mathfrak{g} \quad \forall \mathcal{T} : (\text{Ax} \wedge \text{Fields}_{\mathfrak{g}} \wedge \text{UV}_{\mathfrak{g}} \wedge \text{NT})(\mathcal{T}) \implies \neg(\text{gap} \wedge m < \infty)(\mathcal{T}).$$

The asymmetry between the two statements shapes the whole programme. The positive track must build *one* theory for *each* $\mathfrak{g}$. The negative track must defeat *every* candidate theory for *one* $\mathfrak{g}$, including theories built by methods not yet invented (Section 11).

4 Acceptance contract

Each requirement is atomic, carries its anchor in the normative text and has a class. **NORMATIVE** items restate the mathematical content of S1. **DERIVED** items follow logically from normative ones and are listed separately because they are where attempts typically fail. **PROCEDURAL** items restate S2. **GOVERNANCE** items are recommendations. A candidate resolution is complete only if every **NORMATIVE** and **DERIVED** item is discharged by a theorem in the published work (YM-20–YM-22).

ID	Requirement	Anchor	Class
YM-01	Target object. For each admissible G a quantum field theory $\mathcal{T}_G = (\mathcal{H}, U, \Omega, \mathcal{D}, \{\mathcal{O}_i\}_{i \in I})$ on Minkowski space $\mathbb{R}^{1,3}$ in the sense of Definition 3.1.	S1 §3–4	NORMATIVE
YM-02	All gauge groups. The statement holds for <i>every</i> compact simple gauge group. Constants (including Δ) may depend on G ; uniformity in G is <i>not</i> required. (<i>Reading: I-2.</i>)	S1 §4	NORMATIVE
YM-03	Space-time $\mathbb{R}^4$. The theory lives on $\mathbb{R}^4$: not on a lattice, not on a torus T^4 , not in dimension $d < 4$, and without any remaining UV or IR cut-off.	S1 §4, §6	NORMATIVE
YM-04	Non-triviality. $\mathcal{T}_G$ is non-trivial. The term is not defined in S1; working definition in I-1. (<i>Reading: I-1.</i>)	S1 §4	NORMATIVE
YM-05	Hilbert space and symmetry. A separable Hilbert space $\mathcal{H}$ with a strongly continuous unitary representation U of the restricted Poincaré group $\mathcal{P}_+^\uparrow$ (or its covering group).	S1 §3; S3	NORMATIVE
YM-06	Spectrum condition. The joint spectrum of the generators $(H, \vec{P})$ lies in the closed forward cone $\bar{V}_+$; in particular $H \geq 0$.	S1 §3–4	NORMATIVE
YM-07	Vacuum. A Poincaré-invariant vector $\Omega \in \mathcal{H}$, unique up to a phase.	S1 §3	NORMATIVE
YM-08	Local fields. Operator-valued tempered distributions $\mathcal{O}_i$ on a common dense U -invariant domain $\mathcal{D} \ni \Omega$, Poincaré-covariant, with Ω cyclic for the polynomial field algebra.	S1 §3; S3	NORMATIVE
YM-09	Locality. $[\mathcal{O}_i(f), \mathcal{O}_j(g)] = 0$ on $\mathcal{D}$ whenever $\text{supp } f$ and $\text{supp } g$ are space-like separated.	S1 §3	NORMATIVE
YM-10	Field content. Local fields correspond to gauge-invariant local polynomials in F and its covariant derivatives (e.g. $\text{Tr } F_{ij} F_{kl}$). The correspondence is dimension-filtered, not a naive bijection.	S1 §4 and fn. 1	NORMATIVE
YM-11	Short-distance identity. Correlation functions agree at short distances with asymptotic freedom and perturbative renormalization theory. (<i>Reading: I-4.</i>)	S1 §4	NORMATIVE
YM-12	Stress tensor and OPE. Existence of a stress tensor and of an operator product expansion with the local singularities predicted by asymptotic freedom.	S1 §4	NORMATIVE
YM-13	Mass gap. $\exists \Delta > 0 : \sigma(H) \cap (0, \Delta) = \emptyset$.	S1 §4	NORMATIVE
YM-14	Finite mass. $m := \sup\{\Delta\} < \infty$. Equivalently $H \neq 0$; with A-027 this excludes exactly $\mathcal{H} = \mathbb{C}\Omega$.	S1 §4	NORMATIVE
YM-15	Axiomatic strength. Axiomatic properties at least as strong as S3 and S4. On the Euclidean route: the OS axioms in the corrected form of [OS75], including the linear growth condition (A-001, A-002). (<i>Reading: I-5.</i>)	S1 §4	NORMATIVE
YM-16	No weak existence. Existence by compactness (subsequential limits) alone is excluded unless the axioms and the mass gap are established for the limit.	S1 fn. 2	NORMATIVE

ID	Requirement	Anchor	Class
YM-17	UV regulator removed. Every UV regulator (lattice spacing a , momentum cut-off, ...) is removed; the topology of convergence of the correlation functions is specified.	from YM-03	DERIVED
YM-18	IR regulator removed. Every IR regulator (volume L , boundary conditions) is removed; dependence on boundary conditions is absent or controlled.	from YM-03	DERIVED
YM-19	Properties survive limits. Reflection positivity, full Euclidean invariance $E(4)$ (restored from any lattice subgroup), locality, gauge invariance of observables and the gap bound hold for the <i>limit</i> , not only along the approximation.	from YM-05–16	DERIVED
YM-20	Complete proof. A complete mathematical solution: every step is proved, or reduced to a published theorem whose hypotheses are verified in the instance.	S2 §5(a)	PROCEDURAL
YM-21	Explicit correspondence. The paper addresses the specific questions of the official description; a closely related question does not qualify.	S2 §5(d)	PROCEDURAL
YM-22	Self-containment. CMI does not consider supplementary material submitted by the author: every essential step must be in the published work.	S2 §6(d)	PROCEDURAL
YM-23	Qualifying Outlet. Publication in a refereed mathematics publication of worldwide repute with named editorial board, qualified editors, a published refereeing process and Math-SciNet listing. CMI keeps no list and gives no advice.	S2 §4(a), §6	PROCEDURAL
YM-24	Time and acceptance. At least two years since publication, survival of rigorous examination and general acceptance in the global mathematics community; only then may CMI decide on detailed consideration by a special advisory committee.	S2 §4(b)–(c), §7	PROCEDURAL
YM-25	Priority record. CMI pays special attention to dependence on prior published insights: keep a contribution and priority ledger from day one.	S2 §8(b)	GOVERNANCE

5 Interpretation register

At these points S1 is not formally specific. Each entry records the issue, the working reading of this programme and its consequences for both tracks. Working readings are *hypotheses about CMI’s intent*. S2 states that CMI gives no guidance, so they **MUST** be confirmed through the literature and independent experts (WP-2 in Section 16). An unresolved ambiguity is a candidate for S2 §5(c)(ii) (reformulation), not for the Prize.

I-1 Non-triviality

<i>Issue</i>	S1 requires a “non-trivial” theory but does not define the term.
<i>Working reading</i>	$\mathcal{T}_G$ is not isomorphic to (i) the one-dimensional theory $\mathcal{H} = \mathbb{C}\Omega$ (already excluded by YM-14), (ii) the Wick-polynomial theory of $\dim \mathfrak{g}$ free massless vector fields (already excluded by YM-13, since it is gapless), or (iii) a generalized free (Gaussian) theory. Recommended certificate: prove YM-11 for at least one connected n -point function, $n \geq 3$, whose leading perturbative coefficient is non-zero.
<i>Positive track</i>	The certificate settles YM-04 and part of YM-11 together.
<i>Negative track</i>	A result of the form “only trivial theories exist” must exclude (iii) explicitly.

I-2 Compact simple gauge group

<i>Issue</i>	As an abstract group $SU(N)$ is not simple (centre $\mathbb{Z}_N$). The physics usage is: compact connected Lie group with simple, non-abelian Lie algebra. Different global forms (e.g. $SU(N)$ and $SU(N)/\mathbb{Z}_N$) share the local operator content on $\mathbb{R}^4$ and differ in line operators and θ -periodicity [AST13].
<i>Working reading</i>	The quantifier ranges over all compact simple Lie algebras $\mathfrak{g} \in \{A_n(n \geq 1), B_n(n \geq 2), C_n(n \geq 3), D_n(n \geq 4), E_6, E_7, E_8, F_4, G_2\}$, each with any compact connected G such that $\text{Lie}(G) = \mathfrak{g}$.
<i>Positive track</i>	A method that works only for $SU(3)$, or only for $SU(N)$, does not suffice: all four classical families and all five exceptional algebras are required.
<i>Negative track</i>	A counterexample for one single $\mathfrak{g}$ suffices.

I-3 Theta parameter

<i>Issue</i>	The Lagrangian (1) of S1 has no topological θ -term, although the quantum theory admits one.
<i>Working reading</i>	The target is the $\theta = 0$ theory.
<i>Positive track</i>	$\theta = 0$ suffices (existential quantifier).
<i>Negative track</i>	Unless CMI's reading restricts to $\theta = 0$, a non-existence result would also have to cover $\theta \neq 0$. Flag for expert confirmation.

I-4 Short-distance agreement

<i>Issue</i>	"Agree at short distances with the predictions of asymptotic freedom and perturbative renormalization theory, as described in textbooks" does not specify which correlation functions or to which order.
<i>Working reading</i>	For gauge-invariant local fields, Schwinger functions at non-coinciding points, scaled towards a point, have asymptotic expansions in the running coupling $\bar{g}(\mu)$ that match renormalized perturbation theory order by order. Minimal certificate: leading-logarithmic agreement including the universal coefficients b_0, b_1 (A-009).
<i>Positive track</i>	The order of agreement must be stated explicitly in the main theorem.
<i>Negative track</i>	Ambiguity here is a candidate for S2 §5(c)(ii) (reformulation), not for the Prize.

I-5 Axiom set

<i>Issue</i>	"At least as strong as those cited in [45, 35]": [45] gives the Wightman axioms, [35] is OS I and OS II. The reconstruction claim of OS I is false without the linear growth condition (A-002).
<i>Working reading</i>	Wightman axioms as in Definition 3.1; on the Euclidean route the OS axioms including the linear growth condition of [OS75]. A Haag–Kastler net alone is not sufficient, because the requirement cites [45, 35], not [24].
<i>Positive track</i>	Prove OS0' (linear growth) explicitly; it is the axiom most often forgotten.
<i>Negative track</i>	"The axioms are too restrictive" is an opinion, not a counterexample (Section 11).

I-6 Form of publication

<i>Issue</i>	S2 speaks of "the Proposed Solution" published in a Qualifying Outlet and does not say how a series of papers is treated.
--------------	---

<i>Working reading</i>	A series of refereed papers is acceptable in principle, but one main paper must contain the complete logical chain (the “Clay corollary”) with exact references to the published components.
<i>Positive track</i>	No essential step may live only in code, data or supplements (YM-22).
<i>Negative track</i>	Same rule.

6 Quantifier ledger

Quantifier order is where many apparently complete arguments break. Every estimate in the programme MUST be stored in fully quantified form, and every uniformity it provides MUST be named where it is used.

ID	Topic	Insufficient form	Required form
Q-1	Gauge algebra	$\exists \mathfrak{g}$ (e.g. only $SU(3)$)	$\forall \mathfrak{g} \exists \mathcal{T}_{\mathfrak{g}}$. Constants may depend on $\mathfrak{g}$; uniformity in $\mathfrak{g}$ is <i>not</i> required.
Q-2	Gap along the approximation	$\forall (a, L) \exists c_{a,L} > 0 : \Delta_{a,L} \geq c_{a,L}$	$\exists c > 0, a_0, L_0 \forall a < a_0, L > L_0 : \Delta_{a,L} \geq c \Lambda$
Q-3	Units	bound in lattice units, $a \Delta_{a,L} \geq c$: false as $a \rightarrow 0$ at fixed physical mass	bound in physical units, relative to a renormalized scale Λ fixed by a stated renormalization condition
Q-4	Existence of limits	$\exists$ subsequence $a_k \rightarrow 0$ (compactness; excluded by S1 fn. 2)	full limit, or subsequential limit plus uniqueness, <i>and</i> the axioms and gap proved for the limit
Q-5	Estimates	$\forall a \exists C(a)$	$\exists C \forall a \in (0, a_0)$, with the dependence on L and $\mathfrak{g}$ stated
Q-6	Order of limits	tacit interchange of $a \rightarrow 0$ and $L \rightarrow \infty$	the order (or joint limit) stated, and every interchange proved
Q-7	Counterexample	one construction fails, or one gapless theory exists	$\exists \mathfrak{g} \forall \mathcal{T}$ as in Statement 3.9
Q-8	Clustering	decay for one correlation function	decay for a vacuum-cyclic family, truncated, uniformly (A-016)

7 Assumption ledger

Every definition, imported theorem, perturbative statement, heuristic, numerical fact and working hypothesis the programme touches is recorded here with its type, status, source and scope. Three rules apply:

- R1** No proof step may rest on an entry whose status is **FALSE**, **HEURISTIC**, **NUMERICAL** or **PREPRINT**. `build.mjs` enforces this for all transitions.
- R2** Every **STANDARD** entry MUST be pinned to an exact theorem and page locator before it appears in a manuscript.
- R3** Every entry used in a limit MUST state its uniformity in a , L and $\mathfrak{g}$ (Q-5).

Types: DEF definition; THM theorem or its negation; PERT perturbative statement; HEUR physics heuristic; NUM numerical evidence; HYP working hypothesis (a target theorem); TOOL infrastructure. Entries marked **FALSE** are shortcuts that are known to fail. They are recorded so that nobody uses them by accident.

ID	Statement <i>and scope</i>	Type	Status	Source
A-001	OS reconstruction: Schwinger functions satisfying OS0' (linear growth) and OS1–OS3 define a Wightman theory; cluster (OS4) $\Leftrightarrow$ uniqueness of the vacuum. <i>Exact theorem numbers to be pinned before use.</i>	THM	STANDARD	[OS75]
A-002	“OS axioms without the linear growth condition suffice for reconstruction” (claim of OS I). <i>Proof error (Lemma 8.8 of OS I); corrected in OS II by adding the linear growth condition. Precedent: a foundational claim was falsified and repaired in print.</i>	THM	FALSE	[OS73], [OS75]
A-003	The Wilson lattice gauge theory with compact G is reflection positive. <i>Wilson action; lattice reflections.</i>	THM	VERIFIED	[OSe78]; S1 §6.5
A-004	Lattice Yang–Mills at strong coupling (small β) has exponential clustering (a lattice mass gap) and an area law. <i>Only $\beta < \beta_0(G, d)$. The continuum limit requires $\beta \rightarrow \infty$ (weak coupling): the opposite regime.</i>	THM	PARTIAL	[OSe78], [Sei82], [SZZ23]
A-005	Renormalization-group control of the 4D lattice gauge theory on a torus, uniformly in the lattice spacing (ultraviolet stability). <i>Fixed finite volume. Expectations of gauge-invariant observables and the continuum limit are not completed (S1 §6.5).</i>	THM	PARTIAL	[Bal85]–[Bal89b], [Dim13a], [Dim13b]
A-006	Construction of YM_4 with an infrared cut-off in a continuum regularization. <i>IR cut-off remains; reflection positivity is not automatic for continuum regularizations (S1 §6.5).</i>	THM	PARTIAL	[MRS93]
A-007	“A global continuous gauge-fixing section exists on the space of connections.” <i>Gribov ambiguity; Singer’s theorem. Gauge fixing is only local or patchwise. A failing gauge-fixing method is not a counterexample to S1.</i>	THM	FALSE	[Gri78], [Sin78]
A-008	The formal path integral $Z^{-1}e^{-S_{YM}(A)} \prod_x dA(x)$ is a measure. <i>No such measure is known; guidance only.</i>	HEUR	HEURISTIC	textbooks
A-009	$\beta(g) = -b_0g^3 - b_1g^5 + O(g^7)$ with $b_0 = \frac{11}{3} \frac{C_A}{16\pi^2}$, $b_1 = \frac{34}{3} \frac{C_A^2}{(16\pi^2)^2}$, $C_A = h^\vee(\mathfrak{g}) > 0$ for every simple $\mathfrak{g}$. <i>Perturbative statement. Its non-perturbative meaning must be proved (YM-11).</i>	PERT	VERIFIED	[GW73], [Pol73], [Cas74], [Jon74]
A-010	“The perturbation series in g converges.” <i>Divergent in most known examples; Borel methods of limited use.</i>	PERT	FALSE	S1 §6.1
A-011	Lattice Monte Carlo: lightest 0^{++} glueball of $SU(3)$ at about 1.7 GeV (1730(50)(80) and 1710(50)(80) MeV). <i>Evidence, never proof. Use: sanity check for any claimed rigorous bound.</i>	NUM	NUMERICAL	[MP99], [Chen06]
A-012	“ $\Delta_{a,L} > 0$ for all (a, L) implies a positive gap in the limit.” <i>Non-uniform bounds may tend to 0. A uniform bound in physical units is required (A-031).</i>	THM	FALSE	S1 §5, §6.5
A-013	“A subsequential (compactness) limit settles existence.” <i>Explicitly excluded unless the axioms and the gap are proved for the limit (YM-16).</i>	THM	FALSE	S1 fn. 2
A-014	Scaling limits of critical 4D Ising and lattice ϕ^4 models are Gaussian (marginal triviality). <i>Calibration: a rigorous 4D negative result concerns a specified class of approximations. The published proof itself received a corrigendum (2024).</i>	THM	VERIFIED	[ADC21], [ADC24]

ID	Statement <i>and scope</i>	Type	Status	Source
A-015	Self-interacting single-component ϕ^4 fields do not exist in $d \geq 5$. <i>Calibration for the negative track. Yang–Mills differs: it is asymptotically free.</i>	THM	VERIFIED	[Aiz82], [Frö82]; S1 §6.2
A-016	Under YM-05–YM-09 with a unique vacuum: gap of $H \Leftrightarrow$ gap of $M = \sqrt{H^2 - \vec{P}^2}$ on $\Omega^\perp \Leftrightarrow$ uniform exponential decay in Euclidean time of truncated two-point functions of a vacuum-cyclic field family. <i>Locators to be pinned. Main tool for proving YM-13 from Euclidean data.</i>	THM	STANDARD	[GJ87], [SW64]
A-017	Gap $\Rightarrow$ clustering: for every $C < \Delta$ and every local $\mathcal{O}$ with $\langle \Omega, \mathcal{O}\Omega \rangle = 0$, $ \langle \Omega, \mathcal{O}(\vec{x})\mathcal{O}(\vec{y})\Omega \rangle \leq e^{-C \vec{x}-\vec{y} }$ for large $ \vec{x}-\vec{y} $. <i>Independent consistency check of any gap proof.</i>	THM	VERIFIED	S1 §5
A-018	Full Euclidean invariance $E(4)$ is restored in the continuum limit of the Wilson lattice theory, whose symmetry is only hypercubic. <i>Critical and easy to overlook.</i>	HYP	OPEN	—
A-019	Renormalized gauge-invariant composite fields (e.g. $\text{Tr } F^2$) arise as limits of lattice observables, with finite additive and multiplicative renormalization and mixing only with fields of lower or equal dimension. <i>Needed for YM-10 to YM-12.</i>	HYP	OPEN	S1 fn. 1
A-020	Reflection positivity is preserved under pointwise limits of Schwinger functionals on a fixed class of test functions. <i>Elementary (limits of positive semi-definite forms). The test-function classes along the approximation must match.</i>	THM	STANDARD	[OS73]
A-021	The linear growth condition OS0' holds uniformly along the approximation and passes to the limit. <i>Needed for A-001.</i>	HYP	OPEN	[OS75]
A-022	Confinement (area law for Wilson loops). <i>Not required by S1 (listed as an extension). Must not be used as a premise unless proved.</i>	HEUR	HEURISTIC	S1 §1, §5
A-023	Two-dimensional Yang–Mills on compact surfaces is constructed and exactly solvable. <i>No propagating gluons in 2D: a source of techniques, not evidence for the 4D gap.</i>	THM	VERIFIED	[GKS89], [Dri89], [Wit91], [Lev03]
A-024	The 2D abelian Higgs model satisfies the OS axioms (BFS) and has a mass gap. <i>As of S1, the only complete interacting gauge theory satisfying the axioms.</i>	THM	VERIFIED	[BFS79], [BBIJ84]; S1 §6.4
A-025	The gap may require a non-classical change of variables or duality transformation. <i>Research direction. Any duality used must itself be proved.</i>	HEUR	HEURISTIC	S1 §6.6
A-026	Large- N gauge/string duality for pure Yang–Mills. <i>S1: “no effective approach to the gauge theory without fermions”.</i>	HEUR	HEURISTIC	[tH74]; S1 §6.6
A-027	In a Wightman theory with a unique vacuum, every translation-invariant vector is a multiple of Ω . <i>Locator to be pinned. Used in YM-14 and A-016.</i>	THM	STANDARD	[SW64]
A-028	Compact simple Lie algebras and dual Coxeter numbers: $h^\vee(A_n) = n + 1$, $h^\vee(B_n) = 2n - 1$, $h^\vee(C_n) = n + 1$, $h^\vee(D_n) = 2n - 2$, $h^\vee(E_6, E_7, E_8, F_4, G_2) = 12, 18, 30, 9, 4$. <i>Normalization: long roots of length² = 2.</i>	DEF	VERIFIED	Cartan classification
A-029	Lean 4 formalization of the 4D free field satisfying the Glimm–Jaffe (OS-type) axioms. <i>Tooling precedent for Section 12.</i>	TOOL	PREPRINT	[DHMN26]

ID	Statement <i>and scope</i>	Type	Status	Source
A-030	Public Lean 4 formalization of OS reconstruction (third-party repository). <i>Not independently reviewed here. Candidate infrastructure only.</i>	TOOL	PREPRINT	[OSR26]
A-031	Uniform gap: $\exists c > 0, a_0 > 0, L_0 < \infty$ such that $\Delta_{a,L} \geq c\Lambda$ for all $0 < a < a_0, L > L_0$ (physical units, Λ a renormalized scale). <i>S1: “New ideas are needed”. The core open problem.</i>	HYP	OPEN	S1 §5, §6.5
A-032	Non-triviality certificate of I-1: a connected $n \geq 3$ function of gauge-invariant fields matches its non-zero leading perturbative coefficient. <i>Needed for YM-04.</i>	HYP	OPEN	—
A-033	The construction is uniform in method across all compact simple $\mathfrak{g}$ (or is carried out algebra by algebra). <i>Needed for YM-02. Constants may depend on $\mathfrak{g}$.</i>	HYP	OPEN	—

8 Transition map and closure status

The route from a regularized model to Statement 3.8 consists of separate theorems. Each is listed with the requirements it discharges and the ledger entries it may rest on. The status is *computed*: a transition is **CLOSED** only if all its premises are **VERIFIED** or **STANDARD**. A superscript ! marks a blocking premise.

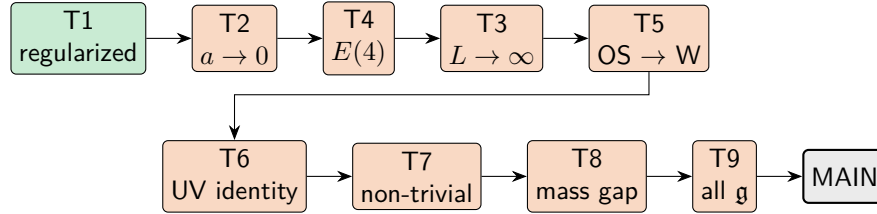


Figure 1: Transition map (one admissible order; T6–T8 may be established in the course of T2–T5).

ID	Transition (separate theorem)	Covers	Premises	Status
T1	Regularized model. Wilson measure $\mu_{a,L}$ on G^{links} : finite-dimensional, gauge invariant, reflection positive.	YM-05	A-003	CLOSED
T2	UV limit $a \rightarrow 0$ at fixed L. Existence of the continuum limit on T^4 for gauge-invariant observables.	YM-17	A-005 [!] , A-019 [!]	OPEN
T3	IR limit $L \rightarrow \infty$. Infinite-volume limit on $\mathbb{R}^4$, driven by a volume-uniform gap or cluster bound.	YM-18, YM-03	A-031 [!]	OPEN
T4	Symmetry restoration. From hypercubic to full $E(4)$ invariance of the limiting Schwinger functions.	YM-19	A-018 [!]	OPEN
T5	Euclidean $\rightarrow$ Minkowski. OS reconstruction of a Wightman theory with unique vacuum.	YM-01, YM-05, YM-06, YM-07, YM-08, YM-09, YM-15	A-001, A-020, A-021 [!]	OPEN

ID	Transition (separate theorem)	Covers	Premises	Status
T6	Field content and UV identity. Renormalized local gauge-invariant fields, asymptotic freedom, stress tensor and OPE.	YM-10, YM-11, YM-12	A-009, A-019 [!]	OPEN
T7	Non-triviality. The limit is not trivial or Gaussian (I-1).	YM-04	A-032 [!]	OPEN
T8	Mass gap in the limit. Positive spectral gap of the reconstructed H ; clustering (A-017) as an independent check.	YM-13, YM-14, YM-16	A-016, A-027, A-031 [!]	OPEN
T9	All gauge groups. T1–T8 for every compact simple $\mathfrak{g}$.	YM-02	A-028, A-033 [!]	OPEN

Failure modes by transition.

- T1 None known. The Wilson measure [Wil74] on a finite lattice is a finite-dimensional integral over copies of the compact group G , it is gauge invariant, and it is reflection positive (A-003). This is why the lattice is the default regularization.
- T2 Controlling the effective actions (A-005) does not by itself control expectations of gauge-invariant observables. Composite fields such as $\text{Tr } F^2$ need additive and multiplicative renormalization and may mix with lower-dimensional fields (A-019). Gauge fixing is only local or patchwise (A-007).
- T3 The gap on a finite torus may close as $L \rightarrow \infty$ (A-012). A volume-uniform bound (A-031) is, per S1 §6.5, where “new ideas are needed”.
- T4 The lattice has only hypercubic symmetry. Rotation invariance of the limit is a theorem to prove, not a convention (A-018).
- T5 Reflection positivity passes to limits (A-020), but the linear growth condition OS0' must hold uniformly (A-021). Without it, reconstruction fails (A-002).
- T6 Agreement with perturbation theory is asymptotic, not convergent (A-010), so the statement proved must be an asymptotic one of explicitly stated order (I-4).
- T7 4D limits can be Gaussian: ϕ_4^4 is the standard warning (A-014). Non-triviality needs a positive certificate (I-1).
- T8 A gap in lattice units is meaningless as $a \rightarrow 0$ (Q-3). The bound must be in physical units and survive both limits.
- T9 A method that uses special properties of $SU(2)$ or $SU(3)$ may not extend to E_8 (I-2).

9 State of the art (verified references)

All entries below were checked against publisher metadata (Crossref) or the primary text. “Missing” names the requirement or transition the result does not reach.

Setting	Result	Missing for MAIN
2D Yang–Mills	Constructed and exactly solvable on compact surfaces [GKS89, Dri89, Wit91, Lev03]	No propagating gluons; not 4D (A-023)

Setting	Result	Missing for MAIN
2D abelian Higgs	OS axioms [BFS79]; mass gap [BBIJ84]	Abelian, 2D. Per S1 the only complete interacting gauge example
3D lattice YM	Ultraviolet stability, uniform in the lattice spacing [Bal85]	Observables, continuum limit, $\mathbb{R}^3$
4D lattice YM on T^4	RG approach with small- and large-field analysis [Bal87, Bal88, Bal89a, Bal89b]; exposition [Dim13a, Dim13b]	T2 incomplete (gauge-invariant expectations); T3–T9 (A-005)
YM ₄ , continuum reg.	Construction with an infrared cut-off [MRS93]	IR cut-off remains; reflection positivity not automatic (A-006)
Lattice, strong coupling	Reflection positivity; exponential clustering and area law at small β [OSe78, Sei82]; stochastic-analysis methods [SZZ23]	Regime opposite to the continuum limit (A-004)
Stochastic quantisation	Langevin dynamics for the 2D YM measure [CCHS22]; 3D Yang–Mills–Higgs [CCHS24]; state space for 3D YM [CC24]	Dimension below 4; reflection positivity and infinite volume not addressed
Lattice, probabilistic	Wilson loops in $\mathbb{Z}_2$ lattice gauge theory [Cha20]; conditional confinement mechanism [Cha21]; correlation decay for finite gauge groups at weak coupling [AC25]	Finite or discrete structure groups, or conditional; no continuum limit
Negative results (scalar)	ϕ_d^4 trivial for $d \geq 5$ [Aiz82, Frö82]; marginal triviality in $d = 4$ [ADC21, ADC24]	Calibration only; not asymptotically free (A-014, A-015)
Numerics	Glueball spectrum of $SU(3)$ [MP99, Chen06]	Evidence, not proof (A-011)
Formal verification	4D free field satisfying OS-type axioms in Lean 4 [DHMN26]; OS reconstruction in Lean 4 (third party) [OSR26]	Preprint or unreviewed; free theory only
Reviews	Status reports [Dou04, Dou26]	—

Claims since 2025. Several preprints and repository deposits claim complete solutions. One example is [Jac25], which addresses the case $G = SU(3)$. As of 2026-09-24, CMI lists the problem as unsolved, and no claim has met the conditions of S2 §4. The example also shows why YM-02 matters: even a correct proof for one gauge group would settle only one instance of the universal quantifier (I-2, Q-1).

10 Track P: construction and mass gap

10.1 Choice of representation

The representation is chosen by which requirements it makes *structurally* safe, and which it leaves to be proved.

Representation	gauge inv.	RP	$E(4)$	AF control	Main risk
R1 Wilson lattice + RG	exact	yes	restore	Balaban	T2 observables, T3, T4
R2 Continuum regularization	gauge-fixed	no	yes	MRS	recovering RP (S1 §6.5)
R3 Stochastic quantization	covariant	no	yes	partial	RP, infinite volume, $d = 4$
R4 Hamiltonian lattice	exact	manifest	restore	weak	Lorentz invariance, UV
R5 New variables / duality	—	—	—	—	the equivalence itself must be proved (A-025)

Recommendation. R1 is the backbone: it is the only representation in which gauge invariance is exact and reflection positivity holds at every cut-off (A-003), which is the bridge to the Hilbert space (S1 §6.5). R2–R5 are auxiliary tools; any result obtained in them MUST be transferred to R1 or the equivalence proved.

10.2 Milestones

- P1 Precise, fully quantified statements of T2–T9 as target theorems. *Exit:* each statement reviewed against the requirements it covers.
- P2 Dependency map of [Bal85]–[Bal89b] and [Dim13a, Dim13b]: what is proved, in which norms, uniformly in what. *Exit:* A-005 split into pinned sub-entries.
- P3 Gauge-invariant observables and renormalized composite fields on T^4 (T2, A-019).
- P4 Rotation invariance of the limit (T4, A-018).
- P5 Volume-uniform gap or cluster estimate (T3, A-031). This is the core open problem.
- P6 $OS0'$ uniformly and reconstruction (T5, A-021).
- P7 Short-distance identity and non-triviality certificate (T6, T7).
- P8 Extension to every $\mathfrak{g}$ (T9) and the Clay corollary (Section 13).

11 Track N: foundational audit and counterexample

11.1 What the “questionable foundation” hypothesis must become

The hypothesis that a widely accepted *premise*, rather than a missing technique, is the obstacle is scientifically legitimate. It has precedents, and it is testable. It becomes testable only in the form of Statement 3.9. For every transition T_k with premises $A_1, \dots, A_n$, the audit asks two separate questions: whether $A_1 \wedge \dots \wedge A_n \Rightarrow T_k$ holds, and whether each A_j is true in the generality in which it is used. Both a proof and a counter-model are attempted, and the outcome is recorded in the ledger.

11.2 Admissible outcomes

- N1 *Non-existence.* For some $\mathfrak{g}$, no $\mathcal{T}$ satisfies $Ax \wedge \text{Fields}_{\mathfrak{g}} \wedge UV_{\mathfrak{g}} \wedge NT$. This is an inconsistency among the normative requirements themselves.
- N2 *Gaplessness.* For some $\mathfrak{g}$, every such $\mathcal{T}$ is gapless. Exhibiting *one* gapless theory refutes nothing, because MAIN is existential in $\mathcal{T}$. N2 therefore requires control over all candidates, in practice a uniqueness or classification theorem.

- N3 *Ill-posedness*. A requirement is shown to be ambiguous or inconsistent in a way that CMI resolves by reformulation. This is the S2 §5(c)(ii) case (a small prize from other CMI funds, not the Prize). The interpretation register (Section 5) is the natural starting point.

11.3 What is not a counterexample

None of the following is a counterexample: failure of a gauge-fixing method (A-007); divergence of perturbation theory (A-010); failure of one regularization or one construction; lattice artefacts; numerical anomalies; the opinion that the axioms are too restrictive (S2 §5(d) excludes papers that address “closely related scientific questions” instead of the specified ones); or a new theory in which the axioms do not hold.

11.4 Precedents: how foundations were actually corrected

Case	What was corrected	Lesson for Track N
OS I → OS II [OS73, OS75]	A reconstruction theorem needed an extra hypothesis (linear growth)	Correction by a precise counterexample to one lemma, published and accepted
Gribov, Singer [Gri78, Sin78]	Global continuous gauge fixing is impossible	Refutes a <i>method</i> , not Yang–Mills theory
Triviality of ϕ^4 [Aiz82, Frö82, ADC21]	Natural lattice approximations give Gaussian limits	Rigorous negative results concern a specified class of approximations
Corrigendum [ADC24]	A published Annals proof needed correction	Even landmark proofs are audited after publication

11.5 Prior assessment

The numerical glueball spectrum (A-011), asymptotic freedom for every $\mathfrak{g}$ (Table 1) and the consistency of the physical picture make outcomes N1 and N2 very unlikely. This is an assessment, not a theorem. The realistic yield of Track N is different: it exposes hidden premises in Track P, and it sharpens or resolves the interpretation register (N3). Track N is pursued because it makes Track P honest, not because a counterexample is expected.

12 Evidence standard and verification

12.1 Evidence classes

Class	Name	May MAIN depend on it?
E1	Proved in the manuscript	Yes
E2	Imported theorem	Yes, if the exact locator is given and every hypothesis is verified in the instance
E3	Machine-checked	Yes, as a complement; the mathematical argument MUST still be in the published text (YM-22)
E4	Numerical evidence	No. Used for discovery and falsification only
E5	Heuristic	No. Used for guidance only

12.2 Adversarial review

Every claimed lemma is given to reviewers with the brief “find the first false statement”, organized by category: quantifier order; limits and their interchange; operator domains and unbounded operators; distributions and test-function classes; analytic continuation; gauge invariance and gauge fixing; composite-operator renormalization; reflection positivity; linear growth $OS0'$; restoration of $E(4)$; UV bounds; IR and volume-uniform bounds; $\mathfrak{g}$ -dependence of constants; spectral limits; non-triviality; circularity.

12.3 Mandatory sanity tests

A proof mechanism MUST fail where the conclusion is false:

- (a) **Abelian control.** For $G = U(1)$ (free Maxwell theory, gapless) the argument must not yield a gap. If it does, it does not use the non-abelian structure and is wrong.
- (b) **Free control.** At $g = 0$ the argument must not yield a gap.
- (c) **Scalar control.** Applied to ϕ_4^4 , the argument must not produce a non-trivial limit in the cases where the limit is proved to be Gaussian [ADC21].
- (d) **Numerical control.** Any rigorous bound $m \geq c\Lambda$ must be consistent with lattice values (A-011) in a matched scheme. A violation is a red flag.
- (e) **Clustering control.** The gap and the decay of correlations (A-017) must agree.

12.4 Formal verification

Formalization in Lean 4 with Mathlib is planned for the parts where hidden hypotheses are most dangerous, in this order: (1) the logical skeleton, i.e. Statement 3.8, Statement 3.9 and the Clay corollary as an implication between abstract structures; (2) inequality chains of the core estimates; (3) combinatorial bounds of cluster and polymer expansions; (4) Lie-algebraic identities (Table 1). Precedents exist [DHMN26, OSR26]. A formal proof shows that every hypothesis is explicit. It does not replace the published argument.

13 Manuscript architecture and traceability

13.1 Architecture

The main theorem appears on pages 1–3. The chapter structure follows the transitions: framework and conventions; imported results (E2, each with its locator); regularized theory (T1); gauge structure; UV control and continuum limit (T2, T4); infinite volume (T3); reconstruction (T5); UV identity (T6); non-triviality (T7); mass gap (T8); all $\mathfrak{g}$ (T9); adversarial checks (the sanity tests above); and the *Clay corollary*: a few lines deriving Statement 3.8 from the preceding theorems.

13.2 Writing rules

- Every theorem is fully quantified. After each estimate, one sentence states which uniformity is used downstream and where (for example: “Independence of L is essential for Proposition 9.2”).
- At critical steps, “clearly”, “obviously” and “one easily sees” are not permitted. This applies in particular to interchanging limits, integrals and sums, domains of unbounded operators, analytic continuation and spectral limits.

- Numerical results appear only in visibly separate “Numerical evidence” boxes.

13.3 Traceability matrix (template)

Requirement	Result in the paper	Evidence	Independent check / reviewer
YM-02	Theorem ...	E1	...
YM-03	Theorems ...	E1/E2	...
YM-13	Theorem ...	E1	clustering check (A-017)
...	...	...	...

Every NORMATIVE and DERIVED requirement MUST have a row. A requirement without a row is an open gap.

14 Publication, priority and integrity

Path set by S2. There is no direct submission to CMI (S2 §5(e)), and CMI gives no advice on venues and keeps no list of them (S2 §6(b)–(c)). The path is: publication in a Qualifying Outlet (YM-23); at least two years of rigorous examination and general acceptance by the global mathematics community (YM-24); then CMI’s decision on detailed consideration by a special advisory committee of at least three members, at least two of them experts in the field (S2 §7(a)(ii)). CMI decisions are final and come without explanation (S2 §3(c)–(d)).

Priority record. From the first working note onward: version control with commit hashes for every manuscript state; an ORCID iD; arXiv versions; and a contribution ledger recording who contributed which idea. The ledger also serves S2 §8(b), under which CMI pays special attention to dependence on prior published insights and may recognize prior work or divide the Prize.

AI assistance. AI tools are not authors. Their use is disclosed, naming the tool, its version and its purpose [COPE23]. Every AI-produced mathematical statement is treated as E5 until a human has proved or verified it at E1–E3.

15 Review of the working draft

The working draft of this programme (German, 2026-09-24, sections numbered 1–33) was checked against S1 and S2. Identifiers in the second column use the draft’s own numbering, not this document’s. Findings are marked **CONFIRMED**, **SHARPENED** or **CORRECTED**.

ID	Draft item	Finding	Detail and source
E1	Intro: Clay counterexample rule	CONFIRMED	Verbatim S2 §5(b)–(c). Addendum: the small prize is paid “from other CMI funds”, at CMI’s sole discretion.
E2	§1, YM-07: axiom level	SHARPENED	S1 says “at least as strong as those cited in [45, 35]”; [35] includes OS II. On the Euclidean route the linear growth condition OS0’ is mandatory. The draft’s §9 list (“Regularität, euklidische Kovarianz, ...”) should name it explicitly (A-002).

ID	Draft item	Finding	Detail and source
E3	§2, YM-02; §21 ch. 13	CORRECTED	“Uniformity in G ” is not required. S1 requires $\forall G \exists \mathcal{T}_G$, and constants may depend on G (Q-1). Requiring uniformity would make the task strictly harder than the problem.
E4	§5 table: “compact simple gauge group”	CORRECTED	The draft treats the term as unambiguous. It is not: abstract simplicity vs. simple Lie algebra, global forms, and θ (I-2, I-3).
E5	§7: two limits	SHARPENED	Missing item: restoration of full $E(4)$ invariance from the hypercubic lattice symmetry (T4, A-018).
E6	§8: uniform gap	SHARPENED	The bound must be in physical units relative to a renormalized scale. In lattice units $a\Delta_{a,L} \rightarrow 0$ is expected (Q-3).
E7	§14: clustering	CORRECTED	S1 §5 states the bound for local $\mathcal{O}$ with $\langle \Omega, \mathcal{O}\Omega \rangle = 0$, at spatial separation. Without this hypothesis the bound is false (Section 3.4).
E8	§15: not required	CONFIRMED	Confinement, chiral symmetry breaking and QCD with quarks are extensions (S1 §1, §5).
E9	YM-17: finite mass	SHARPENED	$m < \infty \iff H \neq 0$, which excludes exactly $\mathcal{H} = \mathbb{C}\Omega$ (Lemma 3.6).
E10	§19, §28: counterexample	SHARPENED	The logical form is $\exists \mathbf{g} \forall \mathcal{T}$. One gapless theory refutes nothing without a uniqueness theorem (N2, Q-7).
E11	§10: RP of the Wilson lattice	CONFIRMED	S1 §6.5, citing [OSe78].
E12	§12, YM-06: operator correspondence	CONFIRMED	S1 fn. 1: not 1–1; the classical differential polynomials of dimension $\leq d$ are expected to correspond to the local quantum operators of dimension $\leq d$.
E13	§29: path to CMI	CONFIRMED	S2 §4, §5(e), §7. Addendum: the special advisory committee has at least three members, at least two of them experts (S2 §7(a)(ii)(3)).
E14	§30: supplements	CONFIRMED	Verbatim S2 §6(d).
E15	§31: division of the Prize	CONFIRMED	S2 §8(a)(iii), §8(b).
E16	Closing: refinement of axioms	SHARPENED	S1 §3 hopes for refinements that <i>add</i> properties (OPE, stress tensor). This does not license weakening the required axioms, which must be “at least as strong”.

16 Roadmap

This is a research programme of open-ended duration. No timeline is promised. Each work package has an exit criterion.

WP	Content	Exit criterion
WP-0	Target freeze, contract, ledgers (this document)	Done in v1.0; build passes
WP-1	Pin every STANDARD entry	No STANDARD entry without theorem and page locator
WP-2	Confirm or revise I-1–I-6	Written opinions of at least two independent experts in constructive QFT
WP-3	Deep reading of the core literature: Balaban’s series and Dimock’s exposition, [MRS93], [OS75], [Sei82]	A-005, A-006 split into pinned sub-entries
WP-4	Lean skeleton of MAIN, its negation and the Clay corollary	Compiles; the logical skeleton contains no sorry
WP-5	Numerical falsification lab (small $SU(2)$ lattices)	Reproduces known gap estimates; all results tagged E4
WP-6	Target theorems T2–T9, fully quantified (P1)	Reviewed against Q-1–Q-8
WP-7	Track N audit of each transition	Every premise either proved, refuted or classified
WP-8	First adversarial review round	Review report per category of Section 12

A Deutsche Zusammenfassung

Dieses Dokument legt überprüfbar fest, was eine Lösung des Millennium-Problems “Yang–Mills Existence and Mass Gap” leisten muss. **Es enthält keine Lösung.** Das Problem ist laut Clay Mathematics Institute ungelöst (Stand 2026-09-24).

- **Zielfixierung:** Die offizielle Problembeschreibung (Jaffe–Witten) und die Preisregeln von 2018 sind wörtlich zitiert und per SHA-256 eingefroren.
- **Abnahmevertrag:** 25 atomare Anforderungen (YM-01 bis YM-25), jede mit Fundstelle im Originaltext.
- **Interpretationsregister:** 6 Stellen, an denen der offizielle Text nicht formal eindeutig ist, etwa “nicht-trivial”, “kompakte einfache Gruppe”, der θ -Parameter und das Kurzdistanzverhalten.
- **Annahmen-Ledger:** 33 klassifizierte Prämissen. 5 verbreitete Abkürzungen sind ausdrücklich als FALSCH markiert.
- **Übergangskarte:** 9 getrennte Teilsätze vom Gitter bis zur Clay-Aussage, derzeit 1 von 9 geschlossen. Das prüft ein Build-Skript maschinell.
- **Zwei Pfade:** Konstruktion mit Mass Gap, sowie ein Grundlagen-Audit mit der exakten logischen Form eines zulässigen Gegenbeispiels ($\exists g \forall T$).
- **Prüfung des Arbeitsentwurfs:** 16 Befunde: 3 Korrekturen, 6 Präzisierungen, 7 Bestätigungen. Die wichtigsten: Gleichmäßigkeit in G wird *nicht* verlangt. Die Clustering-Abschätzung gilt nur für Operatoren mit $\langle \Omega, \mathcal{O}\Omega \rangle = 0$. “Kompakte einfache Eichgruppe” ist mehrdeutig.

B Build and reproducibility

```
node build.mjs           # validate; write generated/*.tex, ledger.json
node build.mjs --check   # fail if generated files are stale
latexmk -pdf spec.tex    # typeset this document
```

The validator rejects duplicate or dangling identifiers and uncovered mathematical requirements. It also rejects transitions resting on inadmissible premises, and it computes the closure status shown in Section 8. Source fingerprints can be re-checked with `sha256sum` against Section 2.

References

- [JW] A. Jaffe, E. Witten, *Quantum Yang–Mills theory*, official problem description, Clay Mathematics Institute (2000); reprinted in J. Carlson, A. Jaffe, A. Wiles (eds.), *The Millennium Prize Problems*, CMI/AMS (2006). <https://www.claymath.org/wp-content/uploads/2022/06/yangmills.pdf>
- [CMI18] Clay Mathematics Institute, *Millennium Prize Description and Rules*, adopted 26 September 2018. https://www.claymath.org/wp-content/uploads/2022/03/millennium_prize_rules_0.pdf
- [SW64] R. F. Streater, A. S. Wightman, *PCT, Spin and Statistics, and All That*, Benjamin, New York (1964).
- [OS73] K. Osterwalder, R. Schrader, Axioms for Euclidean Green’s functions, *Commun. Math. Phys.* 31 (1973) 83–112. doi:10.1007/BF01645738
- [OS75] K. Osterwalder, R. Schrader, Axioms for Euclidean Green’s functions II, *Commun. Math. Phys.* 42 (1975) 281–305. doi:10.1007/BF01608978
- [GJ87] J. Glimm, A. Jaffe, *Quantum Physics: A Functional Integral Point of View*, 2nd ed., Springer (1987).
- [Haag92] R. Haag, *Local Quantum Physics*, Springer (1992).
- [OSe78] K. Osterwalder, E. Seiler, Gauge field theories on a lattice, *Ann. Phys.* 110 (1978) 440–471. doi:10.1016/0003-4916(78)90039-8
- [Sei82] E. Seiler, *Gauge Theories as a Problem of Constructive Quantum Field Theory and Statistical Mechanics*, Lecture Notes in Physics 159, Springer (1982). doi:10.1007/3-540-11559-5
- [Wil74] K. G. Wilson, Confinement of quarks, *Phys. Rev. D* 10 (1974) 2445–2459. doi:10.1103/PhysRevD.10.2445
- [GW73] D. J. Gross, F. Wilczek, Ultraviolet behavior of non-abelian gauge theories, *Phys. Rev. Lett.* 30 (1973) 1343–1346. doi:10.1103/PhysRevLett.30.1343
- [Pol73] H. D. Politzer, Reliable perturbative results for strong interactions?, *Phys. Rev. Lett.* 30 (1973) 1346–1349. doi:10.1103/PhysRevLett.30.1346
- [Cas74] W. E. Caswell, Asymptotic behavior of non-abelian gauge theories to two-loop order, *Phys. Rev. Lett.* 33 (1974) 244–246. doi:10.1103/PhysRevLett.33.244
- [Jon74] D. R. T. Jones, Two-loop diagrams in Yang–Mills theory, *Nucl. Phys. B* 75 (1974) 531–538. doi:10.1016/0550-3213(74)90093-5
- [tH74] G. ’t Hooft, A planar diagram theory for strong interactions, *Nucl. Phys. B* 72 (1974) 461–473.
- [Gri78] V. N. Gribov, Quantization of non-abelian gauge theories, *Nucl. Phys. B* 139 (1978) 1–19. doi:10.1016/0550-3213(78)90175-X
- [Sin78] I. M. Singer, Some remarks on the Gribov ambiguity, *Commun. Math. Phys.* 60 (1978) 7–12. doi:10.1007/BF01609471
- [Bal85] T. Balaban, Ultraviolet stability of three-dimensional lattice pure gauge field theories, *Commun. Math. Phys.* 102 (1985) 255–275.
- [Bal87] T. Balaban, Renormalization group approach to lattice gauge field theories I, *Commun. Math. Phys.* 109 (1987) 249–301.

- [Bal88] T. Balaban, Renormalization group approach to lattice gauge field theories II: cluster expansions, *Commun. Math. Phys.* 116 (1988) 1–22. doi:10.1007/BF01239022
- [Bal89a] T. Balaban, Large field renormalization I: the basic step of the R operation, *Commun. Math. Phys.* 122 (1989) 175–202. doi:10.1007/BF01257412
- [Bal89b] T. Balaban, Large field renormalization II: localization, exponentiation, and bounds for the R operation, *Commun. Math. Phys.* 122 (1989) 355–392. doi:10.1007/BF01238433
- [Dim13a] J. Dimock, The renormalization group according to Balaban I: small fields, *Rev. Math. Phys.* 25 (2013) 1330010. doi:10.1142/S0129055X13300100
- [Dim13b] J. Dimock, The renormalization group according to Balaban II: large fields, *J. Math. Phys.* 54 (2013) 092301. doi:10.1063/1.4821275
- [MRS93] J. Magnen, V. Rivasseau, R. Sénéor, Construction of YM_4 with an infrared cutoff, *Commun. Math. Phys.* 155 (1993) 325–383.
- [BFS79] D. Brydges, J. Fröhlich, E. Seiler, On the construction of quantized gauge fields I–III, *Ann. Phys.* 121 (1979) 227–284; *Commun. Math. Phys.* 71 (1980) 159–205; 79 (1981) 353–399.
- [BBIJ84] T. Balaban, D. Brydges, J. Imbrie, A. Jaffe, The mass gap for Higgs models on a unit lattice, *Ann. Phys.* 158 (1984) 281–319. doi:10.1016/0003-4916(84)90121-0
- [GKS89] L. Gross, C. King, A. Sengupta, Two dimensional Yang–Mills theory via stochastic differential equations, *Ann. Phys.* 194 (1989) 65–112. doi:10.1016/0003-4916(89)90032-8
- [Dri89] B. K. Driver, YM_2 : continuum expectations, lattice convergence, and lassos, *Commun. Math. Phys.* 123 (1989) 575–616. doi:10.1007/BF01218586
- [Wit91] E. Witten, On quantum gauge theories in two dimensions, *Commun. Math. Phys.* 141 (1991) 153–209. doi:10.1007/BF02100009
- [Lev03] T. Lévy, Yang–Mills measure on compact surfaces, *Mem. Amer. Math. Soc.* 166 (2003), no. 790. doi:10.1090/memo/0790
- [Aiz82] M. Aizenman, Geometric analysis of ϕ^4 fields and Ising models, *Commun. Math. Phys.* 86 (1982) 1–48.
- [Frö82] J. Fröhlich, On the triviality of $\lambda\phi_d^4$ theories and the approach to the critical point in $d \geq 4$ dimensions, *Nucl. Phys. B* 200 (1982) 281–296.
- [ADC21] M. Aizenman, H. Duminil-Copin, Marginal triviality of the scaling limits of critical 4D Ising and $\lambda\phi_4^4$ models, *Ann. of Math.* 194 (2021) 163–235. doi:10.4007/annals.2021.194.1.3
- [ADC24] M. Aizenman, H. Duminil-Copin, Corrigendum: Marginal triviality of the scaling limits of critical 4D Ising and ϕ_4^4 models, *Ann. of Math.* 199 (2024). doi:10.4007/annals.2024.199.1.7
- [AST13] O. Aharony, N. Seiberg, Y. Tachikawa, Reading between the lines of four-dimensional gauge theories, *JHEP* 08 (2013) 115. doi:10.1007/JHEP08(2013)115
- [MP99] C. J. Morningstar, M. Peardon, Glueball spectrum from an anisotropic lattice study, *Phys. Rev. D* 60 (1999) 034509. doi:10.1103/PhysRevD.60.034509
- [Chen06] Y. Chen et al., Glueball spectrum and matrix elements on anisotropic lattices, *Phys. Rev. D* 73 (2006) 014516. doi:10.1103/PhysRevD.73.014516
- [Cha20] S. Chatterjee, Wilson loops in Ising lattice gauge theory, *Commun. Math. Phys.* 377 (2020) 307–340. doi:10.1007/s00220-020-03738-9
- [Cha21] S. Chatterjee, A probabilistic mechanism for quark confinement, *Commun. Math. Phys.* 385 (2021) 1007–1039. doi:10.1007/s00220-021-04086-y
- [CC24] S. Cao, S. Chatterjee, A state space for 3D Euclidean Yang–Mills theories, *Commun. Math. Phys.* 405 (2024), art. 3. doi:10.1007/s00220-023-04870-y

- [CCHS22] A. Chandra, I. Chevyrev, M. Hairer, H. Shen, Langevin dynamic for the 2D Yang–Mills measure, *Publ. Math. IHÉS* 136 (2022) 1–147. doi:10.1007/s10240-022-00132-0
- [CCHS24] A. Chandra, I. Chevyrev, M. Hairer, H. Shen, Stochastic quantisation of Yang–Mills–Higgs in 3D, *Invent. Math.* 237 (2024) 541–696. doi:10.1007/s00222-024-01264-2
- [SZZ23] H. Shen, R. Zhu, X. Zhu, A stochastic analysis approach to lattice Yang–Mills at strong coupling, *Commun. Math. Phys.* 400 (2023) 805–851. doi:10.1007/s00220-022-04609-1
- [AC25] A. Adhikari, S. Cao, Correlation decay for finite lattice gauge theories at weak coupling, *Ann. Probab.* 53 (2025). doi:10.1214/24-AOP1702
- [Dou04] M. R. Douglas, Report on the status of the Yang–Mills Millennium Prize problem, Clay Mathematics Institute (2004). https://www.claymath.org/library/annual_report/douglas_quantum_yang_mills.pdf
- [Dou26] M. R. Douglas, The Yang–Mills Millennium problem, *Nat. Rev. Phys.* 8 (2026) 86–97. doi:10.1038/s42254-025-00909-2
- [DHMN26] M. R. Douglas, S. Hoback, A. Mei, R. Nissim, Formalization of QFT, preprint (2026), arXiv:2603.15770.
- [OSR26] OSReconstruction: Lean 4 formalization of the Osterwalder–Schrader reconstruction theorem, repository (accessed 2026-09-24). <https://github.com/xiyin137/OSreconstruction>
- [Jac25] D. C. Jacobsen, A constructive proof of existence and mass gap for pure SU(3) Yang–Mills in four-dimensional space-time, preprint (2025), arXiv:2506.00284. Cited as an example of an unrefereed claim; not assessed here.
- [COPE23] Committee on Publication Ethics, *Authorship and AI tools*, COPE position statement (2023). <https://publicationethics.org/guidance/cope-position/authorship-and-ai-tools>
- [RFC2119] S. Bradner, Key words for use in RFCs to indicate requirement levels, BCP 14, RFC 2119 (1997).
- [RFC8174] B. Leiba, Ambiguity of uppercase vs lowercase in RFC 2119 key words, BCP 14, RFC 8174 (2017).